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Experiment Title : Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

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Experiment No. 5: Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Hub and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

Theory

1. What is a Hub?

A **hub** is a basic networking device used to connect multiple computers or devices within a network, typically in a **Local Area Network (LAN)**. It acts as a central connection point and allows devices to communicate with each other.

A hub operates at the **Physical Layer (Layer 1)** of the OSI model. It does not understand data; it only transmits electrical signals.

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2. Working Principle of a Hub

The working principle of a hub is very simple:

- A hub receives data (electrical signal) from one connected device.
- It **does not check the destination address**.
- It **broadcasts the data to all connected devices**.
- Each device checks whether the data is meant for it.
- Only the intended device accepts the data; others ignore it.

Key points:

- No intelligence or filtering ability
- Data is sent to all ports
- Can cause network congestion
- Works in **half-duplex mode** (devices cannot send and receive at the same time)

Example:
If 5 computers are connected to a hub and Computer A sends data, the hub sends that data to all 4 other computers.

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3. What is a MAC Address?

A **MAC (Media Access Control) address** is a **unique physical address** assigned to a network interface card (NIC) of a device.

Characteristics:

- It is permanent and assigned by the manufacturer.
- Used to identify devices within a local network.
- Consists of **12 hexadecimal digits (48 bits)**.
- Example format: **00:1A:2B:3C:4D:5E**

MAC address works at the **Data Link Layer (Layer 2)** of the OSI model.

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4. What is an IP Address?

An **IP (Internet Protocol) address** is a **logical address** assigned to each device connected to a network.

It helps identify the device and its location on the network.

Types:

- IPv4 (32-bit) → Example: **192.168.1.1**
- IPv6 (128-bit) → Example: **2001:0db8:85a3:0000:0000:8a2e:0370:7334**

IP address works at the **Network Layer (Layer 3)** of the OSI model.

Purpose:

- Enables communication between devices over the internet or network
- Helps in routing data to the correct destination

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5. What is a Local Area Network (LAN)?

A **Local Area Network (LAN)** is a network that connects computers and devices within a **small geographic area**, such as:

- Home
- School
- Office
- College campus

Characteristics:

- High speed
- Low cost
- Limited geographical area
- Allows sharing of resources (printer, files, internet)

Example:
Computers connected in a computer lab form a LAN.

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6. What is Star Topology?

Star topology is a network arrangement where all devices are connected to a **central device** such as a **hub, switch, or router**.

Structure:

- Each device has a separate cable connected to the central device.
- All communication passes through the central device.

Advantages:

- Easy to install and manage
- Failure of one cable does not affect others
- Easy to detect faults

Disadvantages:

- If the central device fails, the entire network stops working
- Requires more cable

Example:
Most modern Ethernet networks use star topology.

Hardware Components Required

- **PC-PT (x3):** End devices acting as network nodes.
- **Hub-PT:** A generic Physical Layer device used to interconnect the nodes.
- **Copper Straight-Through Cables:** The transmission media used for PC-to-Hub connectivity.

Hardware and Connection Setup

- **Network ID:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **PC Naming Convention:**
    - PC 1: **Batch_Roll_A** (e.g., S11_01_A)
    - PC 2: **Batch_Roll_B** (e.g., S11_01_B)
    - PC 3: **Batch_Roll_C** (e.g., S11_01_C)

Procedure

Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select **Network Devices > Hubs > Hub-PT**. Place it in the center.
2. Select **End Devices > PC** and place four units around the Hub.
3. **Naming:** Click on each PC label and rename them following the convention: S11_01_A, S11_01_B, S11_01_C and S11_01_D.

Step 2: Hardware Identification

1. Click on the **Hub** and select the **Physical** tab.
2. Observe the ports. Note that unlike the Switch, a Hub usually has fewer ports and no "intelligence" indicators (like console ports).

Step 3: Physical Cabling

1. Select the **Copper Straight-Through** cable (solid black line).
2. Connect **S11_01_A (FastEthernet 0)** to **Hub (Port 0)**.
3. Repeat for the other three PCs using Hub ports **1, 2** and **3**.
4. **Observation:** Notice that the link lights turn **Green immediately**. Unlike a Switch, a Hub does not run STP (Spanning Tree Protocol), so there is no "Orange" waiting period.

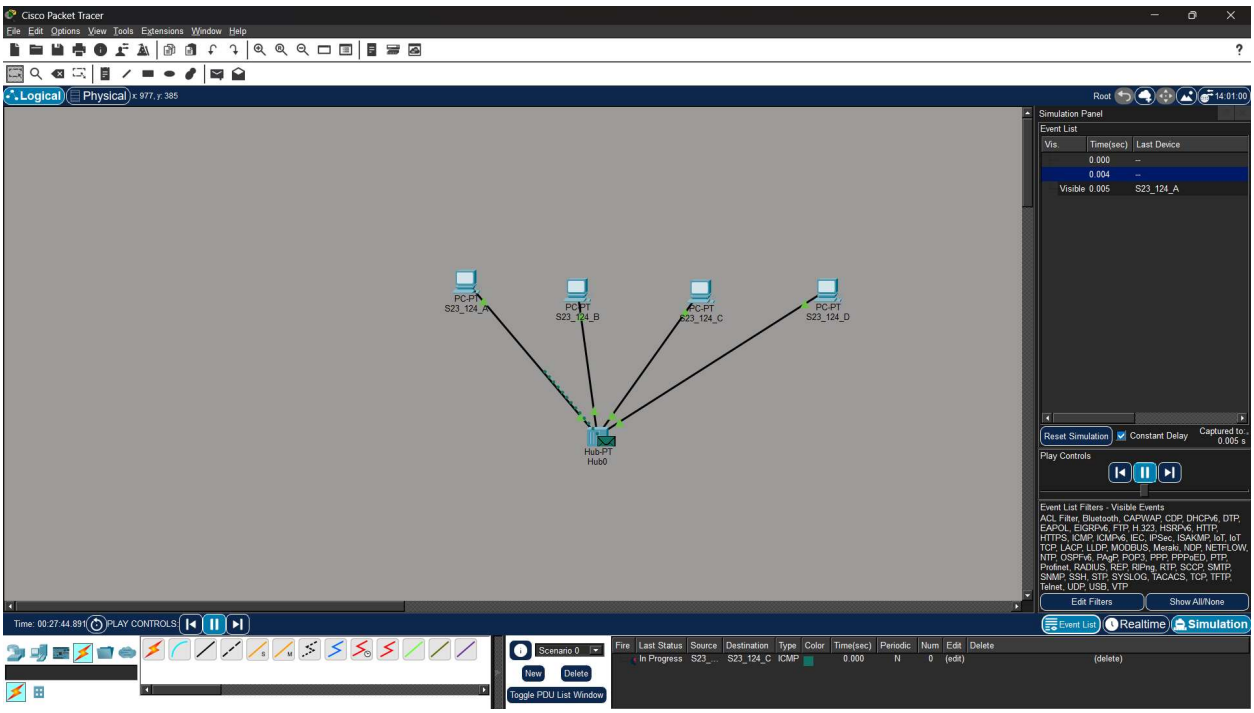
Step 4: Logical Addressing (IP Configuration)

1. Click **S11_01_A > Desktop > IP Configuration**.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure **S11_01_B** with IP 192.168.1.2, **S11_01_C** with IP 192.168.1.3 and **S11_01_D** with IP 192.168.1.4.

Step 5: Testing and Simulation (Crucial Step)

1. **Real-time:** Open the Command Prompt on **S11_01_A** and type ping 192.168.1.3.
2. **Simulation Mode:**
    - Click the **Simulation** tab and filter for **ICMP**.

- Use the **Add Simple PDU** (Envelope) to click **S11_01_A** then **S11_01_C**.
- Click **Play** and observe carefully: The packet goes to the Hub, and the Hub sends it to **all** PC B, PC C and PC D.
- Observe the **Red Cross** on PC B and PC D (because they reject a packet not meant for them) and the **Green Check** on PC C.



6. Observations

- **Instant Connectivity:** Hub ports do not have a "learning" phase; they transition to green immediately upon connection.
- **Broadcast Behavior:** During simulation, the Hub was observed to forward the packet to every connected device. This wastes bandwidth and creates security risks.
- **Device Rejection:** End devices (PCs) are responsible for checking the destination MAC address. If it doesn't match, the PC discards the packet (indicated by the red cross in simulation).

Conclusion

The experiment successfully demonstrates the hardware and functional limitations of a Hub. It concludes that a **Hub** is a non-intelligent Layer 1 device that relies on broadcasting, making it less efficient than a Switch for modern Local Area Networks.